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Cryptography

① Encrypt "CHASE YOUR DREAMS" using :

a. Caesar Cipher

b. Shift Cipher with key 7

c. ROT 13

0	1	2	3	4	5	6	7	8	9
A	B	C	D	E	F	G	H	I	J
10	11	12	13	14	15	16	17	18	19
K	L	M	N	O	P	Q	R	S	T
20	21	22	23	24	25				
U	V	W	X	Y	Z				

→ a. Caesar Cipher : Formula : $C = E(P+3) \bmod 26$

Plain text : CHASE YOUR DREAMS

$$\begin{aligned}C &\rightarrow (P+3) \bmod 26 = (2+3) \bmod 26 = 5 = F \\H &\rightarrow (P+3) \bmod 26 = (7+3) \bmod 26 = 10 = K \\A &\rightarrow (P+3) \bmod 26 = (0+3) \bmod 26 = 3 = D \\S &\rightarrow (P+3) \bmod 26 = (18+3) \bmod 26 = 21 = V \\E &\rightarrow (P+3) \bmod 26 = (4+3) \bmod 26 = 7 = H \\Y &\rightarrow (P+3) \bmod 26 = (24+3) \bmod 26 = 1 = B \\O &\rightarrow (P+3) \bmod 26 = (14+3) \bmod 26 = 17 = R \\U &\rightarrow (P+3) \bmod 26 = (20+3) \bmod 26 = 23 = X \\R &\rightarrow (P+3) \bmod 26 = (17+3) \bmod 26 = 20 = U\end{aligned}$$

$$\begin{aligned}
 D &\rightarrow (P+3) \bmod 26 = (3+3) \bmod 26 = 6 = G \\
 R &\rightarrow (P+3) \bmod 26 = (17+3) \bmod 26 = 20 = U \\
 E &\rightarrow (P+3) \bmod 26 = (4+3) \bmod 26 = 7 = H \\
 A &\rightarrow (P+3) \bmod 26 = (0+3) \bmod 26 = 3 = D \\
 H &\rightarrow (P+3) \bmod 26 = (12+3) \bmod 26 = 15 = P \\
 S &\rightarrow (P+3) \bmod 26 = (18+3) \bmod 26 = 21 = V
 \end{aligned}$$

Ciphertext : FKDVH BRXU GUVDPV

b) Shift Cipher : key : 7

$$\text{Formula : } C = E(P+7) \bmod 26$$

Plain text : CHASE YOUR DREAMS,

$$\begin{aligned}
 C &\rightarrow (P+7) \bmod 26 = (2+7) \bmod 26 = 9 = J \\
 H &\rightarrow (P+7) \bmod 26 = (7+7) \bmod 26 = 14 = O \\
 A &\rightarrow (P+7) \bmod 26 = (0+7) \bmod 26 = 7 = H \\
 S &\rightarrow (P+7) \bmod 26 = (18+7) \bmod 26 = 25 = Z \\
 E &\rightarrow (P+7) \bmod 26 = (4+7) \bmod 26 = 11 = L \\
 Y &\rightarrow (P+7) \bmod 26 = (24+7) \bmod 26 = 5 = F \\
 O &\rightarrow (P+7) \bmod 26 = (14+7) \bmod 26 = 21 = V \\
 U &\rightarrow (P+7) \bmod 26 = (20+7) \bmod 26 = 1 = B \\
 R &\rightarrow (P+7) \bmod 26 = (17+7) \bmod 26 = 24 = Y \\
 D &\rightarrow (P+7) \bmod 26 = (3+7) \bmod 26 = 10 = K \\
 R &\rightarrow (P+7) \bmod 26 = (17+7) \bmod 26 = 24 = Y \\
 E &\rightarrow (P+7) \bmod 26 = (4+7) \bmod 26 = 11 = L \\
 A &\rightarrow (P+7) \bmod 26 = (0+7) \bmod 26 = 7 = H \\
 M &\rightarrow (P+7) \bmod 26 = (12+7) \bmod 26 = 19 = T \\
 S &\rightarrow (P+7) \bmod 26 = (18+7) \bmod 26 = 25 = Z
 \end{aligned}$$

Ciphertext : JOHZL FVBY KYLH TZ

c) ROT 13 : Formula $C = (P + 13) \bmod 26$

Plain text : CHASE YOUR DREAMS

$$\begin{aligned}C &\rightarrow (P + 13) \bmod 26 = (2 + 13) \bmod 26 = 15 = P \\H &\rightarrow (P + 13) \bmod 26 = (7 + 13) \bmod 26 = 20 = U \\A &\rightarrow (P + 13) \bmod 26 = (0 + 13) \bmod 26 = 13 = N \\S &\rightarrow (P + 13) \bmod 26 = (18 + 13) \bmod 26 = 5 = F \\E &\rightarrow (P + 13) \bmod 26 = (4 + 13) \bmod 26 = 17 = R\end{aligned}$$

$$\begin{aligned}Y &\rightarrow (P + 13) \bmod 26 = (24 + 13) \bmod 26 = 11 = L \\O &\rightarrow (P + 13) \bmod 26 = (14 + 13) \bmod 26 = 1 = B \\U &\rightarrow (P + 13) \bmod 26 = (20 + 13) \bmod 26 = 7 = H \\R &\rightarrow (P + 13) \bmod 26 = (17 + 13) \bmod 26 = 4 = E\end{aligned}$$

$$\begin{aligned}D &\rightarrow (P + 13) \bmod 26 = (3 + 13) \bmod 26 = 16 = Q \\R &\rightarrow (P + 13) \bmod 26 = (17 + 13) \bmod 26 = 4 = E \\E &\rightarrow (P + 13) \bmod 26 = (4 + 13) \bmod 26 = 17 = R \\A &\rightarrow (P + 13) \bmod 26 = (0 + 13) \bmod 26 = 13 = N \\M &\rightarrow (P + 13) \bmod 26 = (12 + 13) \bmod 26 = 25 = Z \\S &\rightarrow (P + 13) \bmod 26 = (18 + 13) \bmod 26 = 5 = F\end{aligned}$$

Ciphertext : PUNFR LBHE QERNZF

②

a. "NRNDDOB VHFUHN PHVVDJH"

Using Caesar Cipher.

⇒ Caesar Cipher : Formula : $P = D(C-3) \bmod 26$

Cipher text : NRNDDOB VHFUHN PHVVDJH

$N \rightarrow (C-3) \bmod 26 = (22-3) \bmod 26 = 19 = T$
 $R \rightarrow (C-3) \bmod 26 = (17-3) \bmod 26 = 14 = O$
 $N \rightarrow (C-3) \bmod 26 = (22-3) \bmod 26 = 19 = T$
 $D \rightarrow (C-3) \bmod 26 = (3-3) \bmod 26 = 0 = A$
 $O \rightarrow (C-3) \bmod 26 = (14-3) \bmod 26 = 11 = L$
 $O \rightarrow (C-3) \bmod 26 = (14-3) \bmod 26 = 11 = L$
 $B \rightarrow (C-3) \bmod 26 = (1-3) \bmod 26 = 24 = Y$

$V \rightarrow (C-3) \bmod 26 = (21-3) \bmod 26 = 18 = S$
 $H \rightarrow (C-3) \bmod 26 = (7-3) \bmod 26 = 4 = E$
 $F \rightarrow (C-3) \bmod 26 = (5-3) \bmod 26 = 2 = C$
 $U \rightarrow (C-3) \bmod 26 = (20-3) \bmod 26 = 17 = R$
 $H \rightarrow (C-3) \bmod 26 = (7-3) \bmod 26 = 4 = E$
 $N \rightarrow (C-3) \bmod 26 = (22-3) \bmod 26 = 19 = T$

$P \rightarrow (C-3) \bmod 26 = (15-3) \bmod 26 = 12 = M$
 $H \rightarrow (C-3) \bmod 26 = (7-3) \bmod 26 = 4 = E$
 $V \rightarrow (C-3) \bmod 26 = (21-3) \bmod 26 = 18 = S$
 $V \rightarrow (C-3) \bmod 26 = (21-3) \bmod 26 = 18 = S$
 $D \rightarrow (C-3) \bmod 26 = (3-3) \bmod 26 = 0 = A$
 $J \rightarrow (C-3) \bmod 26 = (9-3) \bmod 26 = 6 = G$
 $H \rightarrow (C-3) \bmod 26 = (7-3) \bmod 26 = 4 = E$

Plain Text : TOTALLY SECRET MESSAGE

"SJB DTNP YNRJ" using shift
Cypher 5

→ Shift cipher: Formula: $P = D(C - 5) \bmod 26$

Cypher text: SJB DTNP YNRJ

$S \rightarrow (C - 5) \bmod 26 = (18 - 5) \bmod 26 = 13 = N$
 $J \rightarrow (C - 5) \bmod 26 = (9 - 5) \bmod 26 = 4 = E$
 $B \rightarrow (C - 5) \bmod 26 = (1 - 5) \bmod 26 = 22 = V$
 $D \rightarrow (C - 5) \bmod 26 = (3 - 5) \bmod 26 = 24 = Y$
 $T \rightarrow (C - 5) \bmod 26 = (19 - 5) \bmod 26 = 14 = O$
 $N \rightarrow (C - 5) \bmod 26 = (22 - 5) \bmod 26 = 17 = R$
 $P \rightarrow (C - 5) \bmod 26 = (15 - 5) \bmod 26 = 10 = K$
 $Y \rightarrow (C - 5) \bmod 26 = (24 - 5) \bmod 26 = 19 = T$
 $N \rightarrow (C - 5) \bmod 26 = (13 - 5) \bmod 26 = 8 = I$
 $R \rightarrow (C - 5) \bmod 26 = (17 - 5) \bmod 26 = 12 = M$
 $J \rightarrow (C - 5) \bmod 26 = (9 - 5) \bmod 26 = 4 = E$

Plain text: NEW YORK TIME

c) "FRAQ ZBER PBSSRR" using ROTB

→ ROTB Formula: $P = D(C + 13) \bmod 26$

Cipher text: FRAQ ZBER PBSSRR.

$$F \rightarrow (C + 13) \bmod 26 = (5 + 13) \bmod 26 = 18 = S$$

$$R \rightarrow (C + 13) \bmod 26 = (17 + 13) \bmod 26 = 4 = E$$

$$A \rightarrow (C + 13) \bmod 26 = (0 + 13) \bmod 26 = 13 = N$$

$$Q \rightarrow (C + 13) \bmod 26 = (16 + 13) \bmod 26 = 3 = D$$

$$Z \rightarrow (C + 13) \bmod 26 = (25 + 13) \bmod 26 = 12 = M$$

$$B \rightarrow (C + 13) \bmod 26 = (1 + 13) \bmod 26 = 14 = O$$

$$E \rightarrow (C + 13) \bmod 26 = (4 + 13) \bmod 26 = 17 = R$$

$$R \rightarrow (C + 13) \bmod 26 = (17 + 13) \bmod 26 = 4 = E$$

$$P \rightarrow (C + 13) \bmod 26 = (15 + 13) \bmod 26 = 2 = C$$

$$B \rightarrow (C + 13) \bmod 26 = (1 + 13) \bmod 26 = 14 = O$$

$$S \rightarrow (C + 13) \bmod 26 = (18 + 13) \bmod 26 = 5 = R$$

$$S \rightarrow (C + 13) \bmod 26 = (18 + 13) \bmod 26 = 5 = R$$

$$R \rightarrow (C + 13) \bmod 26 = (17 + 13) \bmod 26 = 4 = E$$

$$R \rightarrow (C + 13) \bmod 26 = (17 + 13) \bmod 26 = 4 = E$$

Plain text: SEND MORE COFFEE